

Arsalan Ghogari

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GitHub: <https://github.com/arsalanhogari/>

Education

University of Southern California: Viterbi School of Engineering, Marshall

Aug 2023 - Present

School of Business

Bachelor of Science, Computer Science/Business Administration; Minor, AI Applications

Study Abroad: University of Bristol, UK | Spring 2026

- GPA: 3.9; Presidential Scholar, Viterbi Scholar, Dean's List, W.V.T. Rusch Honors, Thematic Option Honors
- Relevant Coursework: Data Structures & Algorithms, Organizational Leadership, Communications, Business Statistics, App Development

Experience

Director of Digital Strategy, USC Undergraduate Student Government

May 2025 - Present

- Manage and maintain USC USG's official WordPress website, improving accessibility, information architecture, and usability for over 20,000 students.
- Developed front-end features using HTML, CSS, and JavaScript, including dynamic homepage announcement systems and interactive election interfaces.
- Built and deployed [Legislative Project Tracker](#) and [Senate Bill Tracker](#) systems by integrating Google Forms, Sheets, and NinjaTables to automate transparency workflows.
- Developed an interactive [Funding Tracker](#) visualizing budget allocation across research, arts, philanthropy, and student initiatives.
- Developing an AI-powered [USC USG Chatbot](#) using OpenAI embeddings, retrieval-augmented generation, and structured website ingestion pipelines to improve student access to campus resources.
- Resolved complex CSV parsing and synchronization issues affecting tracker reliability and maintainability.
- Invited (and accepted) to continue serving in the role for an additional academic year in recognition of technical contributions, initiative, and impact on USC USG digital infrastructure.

Perception & Autonomy Lead / Software Engineer, USC RoboSub

Jan 2024 - Present

- Lead development of the AUV's perception and autonomy stack, integrating computer vision, navigation, SLAM, and mission control for autonomous underwater operation.
- Collected and annotated underwater datasets to train YOLO26 and RT-DETR object detection models, improving detection performance from 0.40 to 0.70 to 0.90 mAP.
- Integrated vision with ROS-based SLAM by publishing perception outputs as ROS topics, enabling real-time localization and navigation using visual detections.
- Developed mission logic and navigation software across the autonomy stack, coordinating perception, planning, and control between ROS nodes.
- Built high-fidelity NVIDIA Isaac Sim environments modeling the AUV, cameras, and sensors with ROS integration, enabling closed-loop sim-to-real testing of the perception and navigation stack prior to in-water deployment.
- Recorded, managed, and analyzed ROS bag datasets in Foxglove to train and debug perception/navigation models, visualize sensor and detection topics, measure latency and detection metrics, and replay runs for reproducible, simulation-driven regression testing.
- Containerized ROS services with Docker and Kubernetes for deployment across NVIDIA Jetson Xavier and Raspberry Pi hardware.
- Built and maintained the team website using React, TypeScript, & Notion API, improving club visibility.

Math Peer Mentor, C.E.N.T.R.I.C. Program (USC Viterbi & Dornsife)

Aug 2025 – Dec 2025

- Supported launch of USC Viterbi and Dornsife's mentoring initiative for first-generation and transfer students through one-on-one mentorship and outreach.
- Collaborated with 30+ professors, advisors, and residential staff to coordinate outreach and increase student engagement.
- Managed mentee application tracking, mentor matching, scheduling, and communication workflows

Technical Projects

[AI-Powered Smart Seasoning Carousel](#) | **React, Tailwind CSS, OpenAI GPT-5.4, ESP32**

- Built an AI-powered smart seasoning carousel integrating an ESP32 microcontroller with a React/Tailwind interface to streamline cooking workflows and ingredient organization.
- Integrated GPT-5.4-based recipe generation to recommend personalized meals and seasoning combinations based on available ingredients and user preferences.

[Collaborative Note-Taking Application](#) | **Java, XML, Android Studio**

- Developed a collaborative Android note-taking application featuring intuitive UI workflows and persistent note management functionality.
- Selected as one of the top 2 implementations out of 37 teams and invited to present the project to the class for technical and design excellence.

[USC USG AI Chatbot](#) | **Python, OpenAI Embeddings, Retrieval-Augmented Generation (RAG), Unstructured AI**

- Developing a retrieval-augmented chatbot to help students access USC campus resources using OpenAI embeddings and structured website ingestion pipelines.
- Designed semantic search workflows to improve information accessibility and reduce friction in navigating university services.

[Personal Portfolio Website](#) | **React, TypeScript, HTML/CSS, JavaScript**

- Designed and developed a responsive personal portfolio showcasing technical projects, leadership experience, and professional achievements with a focus on performance and user experience.
- Built custom interactive components and deployed the site to a custom domain (arsalanhogari.com), serving as a centralized hub for projects, resume, and contact information.

[Study Abroad Diaries \(3D Web Experience\)](#) | **HTML, CSS, JavaScript**

- Developed an interactive 3D photo carousel documenting experiences from studying abroad at the University of Bristol, featuring draggable animations and immersive storytelling.
- Engineered a custom JavaScript carousel with dynamic image rendering, responsive design, and accompanying narratives to create an engaging visual portfolio experience.

Skills

Programming: Python, C++, C, Java, JavaScript, TypeScript, React, HTML/CSS, MATLAB

Technologies & Frameworks: ROS, Kubernetes, Docker, OpenAI API, YOLO, WordPress, Java Servlets

Tools: Git, Google Workspace, MS Office, OnShape, AutoCAD